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University of Information Technology & Sciences

Lab Report-01

Drawing an Entity-Relationship Diagram

Course Title: Software Engineering and System Analysis Lab

Course Code: CSE 356

Submitted To

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Experiment No: 01

Experiment Name: Study of designing Entity-Relationship model

Objectives:

- To identify and list all relevant entities within the context of the database.
- To identify and represent entities, attributes, and relationships within a system.
- To establish a clear understanding of the data elements that need to be represented in the ER model.
- To determine the cardinality and participation constraints for each relationship.

Apparatus Required:

- Computer with internet access.
- ER diagram modeling tool (ERDPlus)

Theory:

An Entity-Relationship (ER) diagram is a visual representation of the data model that defines the entities involved in a system and the relationships between them. It's a visual tool used in database design to illustrate the structure of a database and the way data is related within it. Any object, for example, entities, attribute of an entity, relationship sets, and attribute of relationship sets, can be represented with the help of an ER diagram.

Entity: Entities are represented by means of rectangles. Rectangles are named with the entity set they represented.

Attributes:

Entities are denoted utilizing their properties, known as attributes. All attributes have values. For example, a student entity may have name, class, and age as attributes. There exists a domain or range of values that can be assigned to attributes. For example, a student's name cannot be a numeric value. It has to be alphabetic. A student's age cannot be negative, etc.

There are five types of Attributes:

- Key attribute
- Composite attribute
- Single-valued attribute
- Multi-valued attribute
- Derived attribute

Composite Attribute: Can be subdivided into smaller components.

Single-Valued Attribute: Holds only one value for a given entity.

Multi-Valued Attribute: Can hold multiple values for a single entity.

Derived Attribute: Its value can be derived from other attributes

Key Attribute: Uniquely identifies an entity within an entity set. Example: Student_ID for a student.

Relationship:

A relationship illustrates how two or more entities are associated with each other.

Degree of a relationship set

Unary: It is a relationship between the instances of one entity type.

Binary: It is a relationship between the instances of two entity type.

Ternary: It is a relationship between the instances of three entity type.

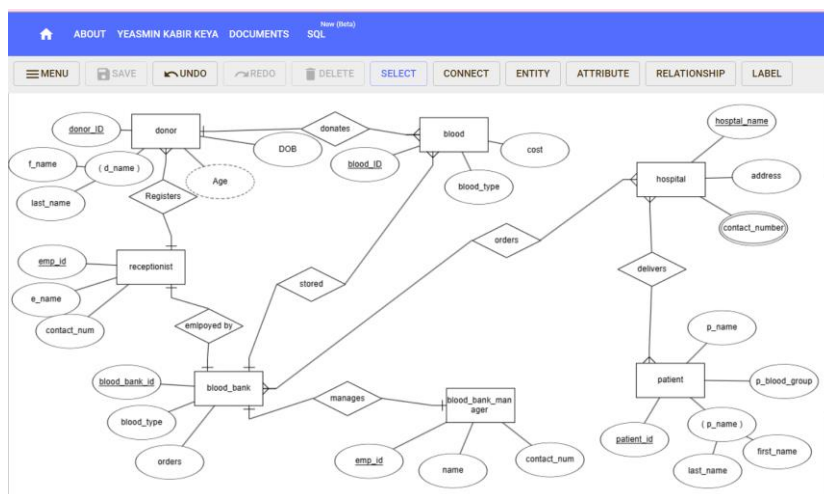
Cardinality:

Cardinality in an ER diagram defines the number of instances of one entity that can be associated with instances of another entity. It describes the relationship constraints between entities.

Types of Cardinality:

1. **One-to-One (1:1):** Each entity in Set A is related to only one entity in Set B, and vice versa.
2. **One-to-Many (1:M):** One entity in Set A can relate to multiple entities in Set B, but each entity in Set B relates to only one entity in Set A.
3. **Many-to-Many (M:M):** Multiple entities in Set A can be associated with multiple entities in Set B.
4. **Many-to-One (M:1):** In a **Many-to-One (M:1) Relationship**, multiple instances of one entity are associated with a single instance of another entity.

Diagram:



Result and Discussion:

The ER diagram effectively captures the essential entities and their relationships within the system. By accurately identifying entities, attributes, and relationships, the diagram serves as a foundational blueprint for database design, ensuring data integrity and efficient data retrieval. This visual representation aids in understanding the system's data requirements and guides the development of a structured and normalized database schema.

Conclusion:

Designing an ER diagram is a crucial step in the software development process. It facilitates a clear understanding of the data components and their interconnections within a system. Mastering ER diagramming techniques enhances one's ability to model complex databases effectively, leading to more robust and maintainable software solutions.